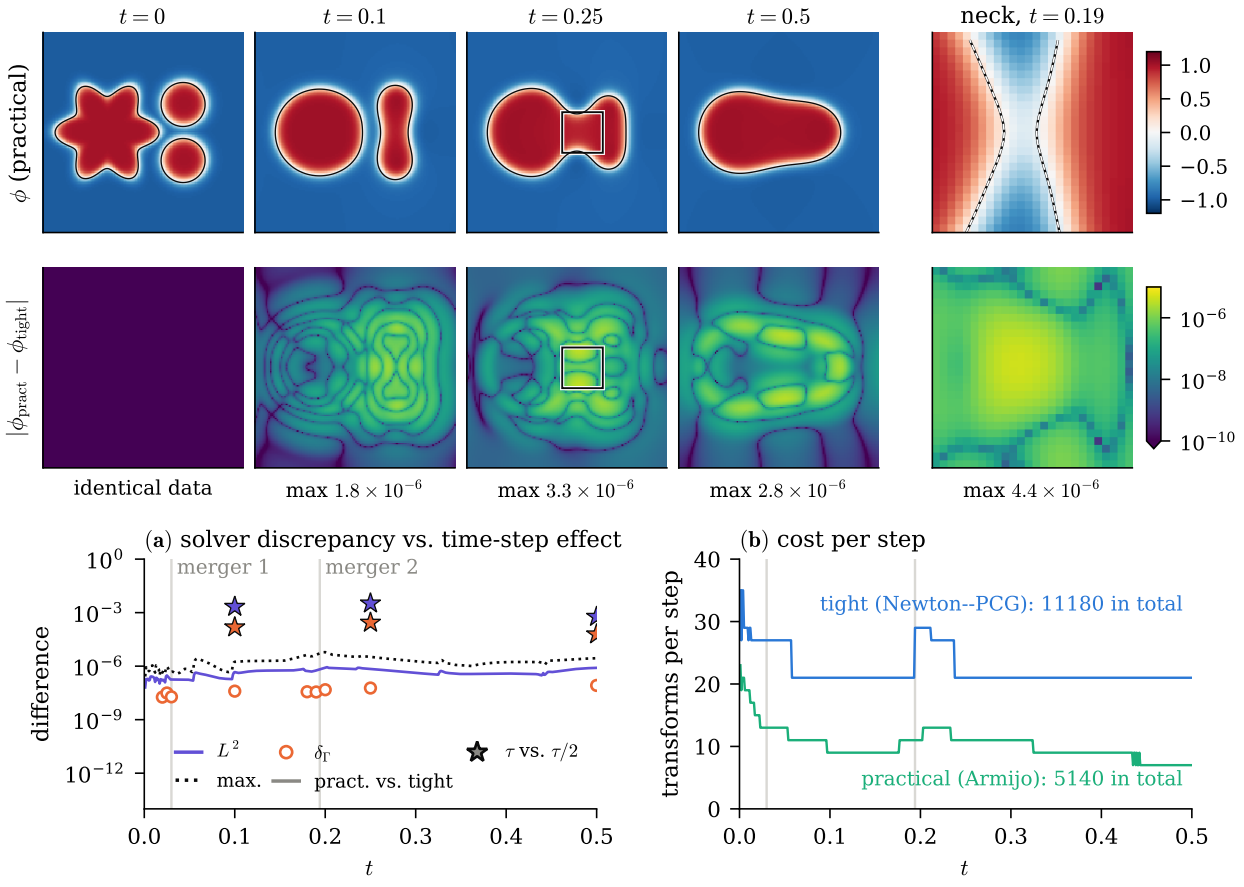

$t = 0$
$t = 0.1$
$t = 0.25$
$t = 0.5$
neck, $t = 0.19$
ϕ (practical)
1.0
0.5
0.0
−0.5
−1.0
$|\phi_{\text{pract}} - \phi_{\text{tight}}|$
10^{-6}
10^{-8}
10^{-10}
identical data
max 1.8×10^{-6}
max 3.3×10^{-6}
max 2.8×10^{-6}
max 4.4×10^{-6}
(a) solver discrepancy vs. time-step effect
merger 1
merger 2
difference
10^{0}
10^{-3}
10^{-6}
10^{-9}
10^{-12}
L^2
δ_Γ
τ vs. $\tau/2$
max.
pract. vs. tight
0.0
0.1
0.2
0.3
0.4
0.5
t
(b) cost per step
transforms per step
40
30
20
10
0
tight (Newton--PCG): 11180 in total
practical (Armijo): 5140 in total
0.0
0.1
0.2
0.3
0.4
0.5
t